Introduction to Data Science - Spring 1405

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Lectures Sunday, Tuesday: 15:00-16:30, Class 5

Syllabus

- Introduction – Data science lifecycle
- Sampling, Sampling Bias, Observational and Experimental Studies, Confounding Variable
- Statistical Charts - Visualizing numerical data
- A review of probability theory, conditional probability, Bayes theorem, random variable, expectation and variance, Normal, binomial, Monte Carlo Simulation
- Central limit theorem, confidence interval, hypothesis testing, power of a test,
- Data Visualization Design Principles
- Data preprocessing, EDA
- SQL, NoSQL, Graph DBs,Mongo DB, BigData, Hadoop
- BigData, Spark, Kafka, HBase, Cassandra
- Feature Engineering
- Modeling Mindset (prediction vs explanation, data splits)
- Evaluation (regression, classification and clustering)

- Linear & Logistic regression
- Decision Trees, Random forests, Boosting, SVMs & KNNs
- MLOps, Kubernetes، Docker, MLflow, Managing Drift & Retraining
- Unsupervised Learning (clustering, dimensionality reduction, visualization vs modeling)
- Neural networks, Feedforward, CNNs & RNNs
- Image classification
- Time Series
- NLP, Embedding, LLMs & Transformers
- RAGs
- Agents
- Interpretability & Explainability
- Privacy, Ethics

Announcements Official announcements are made through eLearn **only**.

Calendar The course calendar is accessible on the course page on eLearn. Please note that the calendar is subject to change and may change throughout the semester.

Telegram Group https://t.me/+4Zzfo8AYmSYxYzVk

Grading Your final grade out of 20 is calculated as follows **(subject to change):**

- 6*1.5 for assignments

- 5 for project
- 6 for final exam
- 1 for contest

Exams

One final exam is scheduled at the times announced in the course calendar. Please note that according to the university law, your presence at the final exam is **mandatory** for the course to be recorded in your transcript.

Assignments

There will be 6 assignments and a project throughout the course. You can do them **individually** or in a group of **maximum 3 students**. If done in a group, each of the members should be able to answer the questions asked about that assignment, or their score will decrease. You can change your group for each assignment, but all of the members for the assignment should be specified when it is about to be delivered.

Late Policy

- For every 1 day delay, there will be a 10% penalty on the Assignment final score.
- You will have a total of 8 days grace period.
- The final project will not be accepted after its initial deadline i.e. there will be no late and grace policy!

Cheating Policy

Your work is considered cheating if:

- You use the ready-made code available on the internet as a solution.
- You use someone else's code.
- You mistakenly upload someone else's code instead of your own!

- Copy the answers generated by Chat GPT without understanding

Penalties for cheating, especially to prevent the spread of cheating, are considered in this course as follows:

- Cheating in an assignment results in a 100% negative score for that assignment for each student involved, whether they are the cheater or the cheated upon;
- Cheating may be detected at any time during the term. If you have once performed an unacceptable act and have not been caught immediately, it does not mean that this check will never occur.
- Cheating in exams is subject to educational regulations and may result in the subject being referred to the educational committee and ultimately the disciplinary committee.

Resources

- Github Page
- Previous Semester Github Page

Environment

Jupyter Notebook/JupyterLab

Staff:

Mohammad Amanlou

(Mohammad Amanlou | LinkedIn)

Co-Chief Teaching Assistant

Mostafa Kermani Nia

(Mostafa Kermani Nia | LinkedIn)

Co-Chief Teaching Assistant

Amir Naddaf Fahmideh

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Teaching Assistant

Mina Shirazi

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Behrad Binaei

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Teaching Assistant

Roshak Masoudi nejad

(Roshak Masoudi Nejad | Github)

Teaching Assistant

Mehrad Liviyan

(Mehrad Liviyan | LinkedIn)

Teaching Assistant

Erfan Falahati

(Erfan Falahati | Github)

Teaching Assistant

Sepanta Ghonoodi

(Sepanta Ghonoodi | LinkedIn)

Teaching Assistant

Maryamm Vali

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Faezeh Mozaffari

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